Department of Computer and Software Engineering
College of Electrical and Mechanical Engineering
National University of Sciences & Technology
Peshawar Road, Rawalpindi – 46000, Pakistan
Website www.ceme.edu.pk Tel: +92-51-54444100 Fax: +92-51-9247548

**TO WHOM IT MAY CONCERN**

I am pleased to write this letter of recommendation for **Mr. Muhammad Hamza Khan**, an outstanding student of the **Master's in Software Engineering** program at the **National University of Sciences and Technology (NUST)**. I have had the privilege of supervising Hamza in both academic and research contexts and can confidently attest to his exceptional technical capabilities, research acumen, and professionalism.

Hamza is currently pursuing a cutting-edge thesis titled: **"A Multi-Agentic Framework for Vision and Language Models Targeting Geospatial Intelligence Using Large Language Models (LLMs)"**, which I co-supervise. His work explores the integration of **LLMs, Vision Transformers, and agentic pipelines** to address challenges in **GeoAI, autonomous spatial systems**, and GIS analysis. His innovative and interdisciplinary approach positions his research at the forefront of intelligent geospatial solutions.

In my **Deep Learning** course, Hamza ranked among the top performers and completed an advanced project on **fingerprint image enhancement** using **Gabor filters, adaptive histogram equalization, and CNN-based denoising**, showcasing his strengths in image processing, pattern recognition, and AI security.

In parallel with his academic excellence, Hamza is a **full-stack AI developer and entrepreneur**, leading **DevZona**, an international AI-focused software lab. He has successfully delivered over **100+ large-scale projects**, including:

- Real-time **Facial Emotion Detection** systems
- **Document Screening Platforms** using LLMs & Vector DBs
- **Multi-Tenant SaaS Applications** with advanced automation
- **AI-based GIS & Remote Sensing Systems**
- **Blockchain-integrated platforms**
- Intelligent **AI Bots** for real estate, content generation, and automation
- **LLM-driven Educational Platforms** for personalized learning

Hamza's technical stack includes **React.js, Next.js, Node.js, Python, Django, TypeScript, MongoDB, PostgreSQL, LangChain, Pinecone, TensorFlow, Keras, Docker, AWS, GCP, LLMs, LLaMA, and OpenAI models**. His ability to combine research with practical implementation is remarkable. Please feel free to contact me for any further information.

Muhammad Usman Akram, PhD
Professor
College of E&ME (NUST)
Rawalpindi, Pakistan

College of Electrical & Mechanical Engineering
DC&SE

**Dr Muhammad Usman Akram**

Professor/ Head of Department
Computer & Software Engineering Department
NUST College of EME
usman.akram@ceme.nust.edu.pk
+923336913921
Director, BIOMISA Lab (www.biomisa.org)
Co-Founder and Director RISETech Pvt Ltd (www.risetech.ai)
https://ceme.nust.edu.pk/faculty/muhammad-usman-akram/